

Project Proposal – Weekly Spotify Recap

Project Overview

I want to solve the daily problem of not having a simple, automatic way to see what music I listened to most and how much time I spent listening overall. At the moment, this information is spread across different apps and is not available as a clear weekly summary. My automated script will collect my listening data, calculate statistics such as top artists, top tracks and total listening time and then generate a weekly recap. It will also create or update a playlist with my top songs and send me an email summary with the results and a link to the playlist so that I can easily listen to my favorite songs.

Key Features

- Automatic collection of weekly music listening data
- Detection of top artists and top tracks of the week
- Calculation of overall listening statistics such as total listening time
- Automatic creation/update of a playlist with the top songs of the week
- Sending an email recap with the weekly summary and playlist link

Must requirements

- Collect weekly listening data automatically
- Identify top songs/albums/artists of the week
- Calculate overall listening statistics
- Create a playlist with the top songs
- Send an email summary with the recap results and playlist link

Could requirements

- Include album covers or artist images in the email
- Save past weekly recaps for later comparison
- Add extra statistics such as most played album, latest listen or number of songs played
- Add clickable links for songs, albums, and artists that open the corresponding Spotify page

Should requirements

- Use an email design inspired by Spotify
- Show rankings and listening time for artists and tracks
- Display the playlist link clearly in the email
- Handle errors properly if data cannot be loaded

Won't have requirement

- No database integration
- No support for multiple music platforms such as Apple Music or others.

Systems Involved

- [Last.fm API](#) (for summaries and highlights)
- [Spotify For Developers API](#) (for top rankings and icons)
- SMTP-Server for Sending out Mails. (weeklyspotifysummary@gmail.com)
- Windows Task-Scheduler for sending weekly emails.
- Local: Python script
- Remote: GitHub for backup

Implementation Steps

1. **Requirements:** Define the project goal and required features, such as collecting weekly listening data, identifying top artists and tracks, calculating overall statistics, creating/updating a playlist and sending an email recap.
2. **Design:** Plan how the script will display the music data and rank it, generate the playlist and format the email summary.
3. **Implementation:** Develop the script so it can collect the data from the APIs, calculate the weekly recap, create the playlist and send the email.
4. **Testing:** Test the script with different listening data and possible error cases to make sure the recap, playlist and email works correctly.
5. **Deployment:** Prepare the script for regularly weekly use with Task Scheduler (e.g. every Monday at 9 AM)
6. **Documentation:** Document the project steps, explain how the script works, reflect the project and note what was good and what could've done better and describe how to set it up and use it.

Benefits

- Time savings because the weekly recap is created automatically
- Clear overview of personal listening habits
- Automatic playlist creation with the favorite songs of the week
- Easy access to results through a structured email recap
- Reduced manual effort when checking and summarizing music statistics